

21) a) Use equation (2) from Section 2.1 to show that $\text{row}_i(A) = \text{row}_i(I) \cdot A$, for $i=1,2,3$.

Solution: $\text{row}_i(A) = \text{row}_i(I \cdot A)$ (Theorem 2(e))
 $= \text{row}_i(I) \cdot A$ (Equation 2 from Sec. 2.1) $\square$
 $i=1,2,3$

b) Show that if rows 1 and 2 of A are interchanged, then the result may be written as EA , where E is an elementary matrix formed by interchanging rows 1 and 2 of I .

Solution: ~~$\text{row}_1(A)$~~ ~~$\text{row}_2(A)$~~ $\text{new } A = \begin{bmatrix} \text{row}_2(A) \\ \text{row}_1(A) \\ \text{row}_3(A) \end{bmatrix} = \begin{bmatrix} \text{row}_2(I) \cdot A \\ \text{row}_1(I) \cdot A \\ \text{row}_3(I) \cdot A \end{bmatrix}$ (part a)
 $= \begin{bmatrix} \text{row}_2(I) \\ \text{row}_1(I) \\ \text{row}_3(I) \end{bmatrix} \cdot A$ (Eqn. 2)
 $\therefore E = \begin{bmatrix} \text{row}_2(I) \\ \text{row}_1(I) \\ \text{row}_3(I) \end{bmatrix}$ which is an elementary matrix formed by

interchanging rows 1 and 2 of I . $\square$

c) Show that if row 3 of A is multiplied by 5, then the result may be written as EA , where E is formed by multiplying row 3 of I by 5.

Solution: $\text{new } A = \begin{bmatrix} \text{row}_1(A) \\ \text{row}_2(A) \\ 5 \cdot \text{row}_3(A) \end{bmatrix} = \begin{bmatrix} \text{row}_1(I) \cdot A \\ \text{row}_2(I) \cdot A \\ 5 \cdot \text{row}_3(I) \cdot A \end{bmatrix}$ (Part a)
 $= \begin{bmatrix} \text{row}_1(I) \cdot A \\ \text{row}_2(I) \cdot A \\ (5 \cdot \text{row}_3(I)) \cdot A \end{bmatrix}$ (Theorem 2(d)) $= \begin{bmatrix} \text{row}_1(I) \\ \text{row}_2(I) \\ 5 \cdot \text{row}_3(I) \end{bmatrix} \cdot A$ (Eqn. 2 from Sec. 2.1)
 $\therefore E = \begin{bmatrix} \text{row}_1(I) \\ \text{row}_2(I) \\ 5 \cdot \text{row}_3(I) \end{bmatrix}$. The matrix is formed by multiplying row 3 of I by 5. $\square$

28) Show that if row 3 of A is replaced by $\text{row}_3(A) - 4 \cdot \text{row}_1(A)$, the result is EA , where E is formed from I by replacing $\text{row}_3(I)$ by $\text{row}_3(I) - 4 \cdot \text{row}_1(I)$.

Solution: $\text{new } A = \begin{bmatrix} \text{row}_1(A) \\ \text{row}_2(A) \\ \text{row}_3(A) - 4 \cdot \text{row}_1(A) \end{bmatrix} = \begin{bmatrix} \text{row}_1(I) \cdot A \\ \text{row}_2(I) \cdot A \\ \text{row}_3(I) \cdot A - 4(\text{row}_1(I) \cdot A) \end{bmatrix}$ (Part a)